LECTURE / SESSION[1] PLAN

Name Lecturer: Sofia Gil-Clavel	**Date session:** 1	**Expected number of students:** 20
Course title	R tools for teaching and research	
Topic of lecture or session	Recap R and tidyverse	
Situational factors (e.g. group size, prior knowledge, expected motivation)	The class lasts 3hrs. The students have at least basic knowledge of R and tidyverse. The students use different Operating Systems (Windows, Mac, or Linux).	
Intended learning outcomes of this session	At the end of the session the students will have refreshed their memories in regards to: 1. Explain the difference between R and RStudio. 2. Explain the difference between base R and tidyverse. 3. Explain where to look for help. 4. Use tidyverse for data manipulation. 5. Use tidyverse for data visualization. 6. Use base R for data modeling.	
Learning material (book, chapters, …)	The lecture is based on the book: • Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. R for data science. " O'Reilly Media, Inc.", 2023. Accessed May 7, 2024. https://r4ds.hadley.nz/.	
Media, equipment, tools	The students use their own laptops. The teacher needs access to a projector and a whiteboard.	

[1] A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

Preparation for students	The students have access to the slides and codes before the class: https://github.com/SofiaG1l/Course_Rtools/1_RecapRandTidyverse

Time[2] (min.)	**Didactic element (goal)[3] and topic[4]**	**What the teacher does[5]** (teacher activity)	**What students do** (learner activity)	**Evaluation[6]** (feedback/assessment)
10	• Explain the difference between R and RStudio.	• The teacher uses the slides to remind the students what R is. • The teacher uses the slides to remind the students what RStudio is. • The teacher explains the different panels in RStudio.	• Before telling each element the students learned, the teacher waits some seconds for the students to fill out the information out loud. • The students passively digest what the teacher is explaining. • The students open RStudio and follow the steps that the teacher is explaining.	• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.
10	• Explain the difference between base R and tidyverse.	• The teacher uses the slides to remind the students what base R is. • The teacher uses the slides to remind the students what a package is and how to install them and use them	• The students passively digest what the teacher is explaining. • The students open and follow the steps that the teacher is explaining.	• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.

[2] Indicate the planned duration in minutes
[3] State the number(s) of the relevant ILO at each didactic element
[4] Only key words
[5] Specify the type of activity and write down also the questions that you prepared to ask in the session
[6] Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

		• The teacher uses the slides to remind the students what tidyverse is.		• The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
50	• Use tidyverse for data manipulation.	• The teacher uses the slides to remind the students how to use the tidyverse packages: tibble, pipe, and dplyr. • The teacher asks the students to solve a series of data science exercises using tidyverse.	• The students passively digest what the teacher is explaining. • The students solve the exercises in their computers. • At the end the students and the teacher solve the exercises together.	• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student. • The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
50	• Use tidyverse for data visualization.	• The teacher uses the slides to remind the students what ggplot2 is.	• The students passively digest what the teacher is explaining. • The students solve the exercises in their computers.	• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these

		• The teacher asks the students to solve a series of exercises where they visualize data.	• At the end the students and the teacher solve the exercises together.	confusions the teacher will be able to correct the student. • The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
50	• Use base R for data modeling.	• The teacher reminds the students how to use base R for data modeling. • The students solve a series of exercises to model some data.	• The students passively digest what the teacher is explaining. • The students solve the exercises in their computers. • At the end the students and the teacher solve the exercises together.	• The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher performs the step again in her computer.
10	• Explain where to look for help.	• The teacher uses the slides to explain the "help" panel and the Cheat Sheets.	• The students perform an exercise where they learn more about pipes by following the Vignette example from the package.	• The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.